

# CPSC-406 Report

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## Abstract

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## 1 Introduction

## 2 Week by Week

### 2.1 Week 1

#### Exercise 1

Consider the following DFAs  $\mathcal{A}_1$  and  $\mathcal{A}_2$ . Complete the following table with all 8 possible strings of length 3 over  $\{a, b\}$  (columns: Accepted by  $\mathcal{A}_1$ ? and Accepted by  $\mathcal{A}_2$ ?).

**Answer to Exercise 1.1.** Here is my completed table (words of length 3 over  $\{a, b\}$ ):

| word       | Accepted by $\mathcal{A}_1$ ? | Accepted by $\mathcal{A}_2$ ? |
|------------|-------------------------------|-------------------------------|
| <i>aaa</i> | No                            | Yes                           |
| <i>aab</i> | Yes                           | No                            |
| <i>aba</i> | No                            | No                            |
| <i>abb</i> | No                            | No                            |
| <i>baa</i> | No                            | Yes                           |
| <i>bab</i> | No                            | No                            |
| <i>bba</i> | No                            | No                            |
| <i>bbb</i> | No                            | No                            |

### Exercise 1.2

Describe the languages  $L(\mathcal{A}_k)$  accepted by  $\mathcal{A}_k$  for  $k = 1, 2$ .

**Answer to Exercise 1.2.** In both cases the language is the same: for  $k = 1, 2$ ,  $L(\mathcal{A}_k)$  is *all the words over  $\{a, b\}$  that end with  $ab$* .

### Exercise 2 (Designing DFAs)

Design DFAs whose accepted languages are given as follows:

1. All the words that end with  $ab$
2. All the words that contain  $aba$
3. All the words that contain an odd number of  $a$ 's and an odd number of  $b$ 's
4. All the words that contain an even number of  $a$ 's and an odd number of  $b$ 's
5. All the words such that any three consecutive characters contain at least one  $a$
6. All the words that contain  $bbb$

**What do you notice when comparing the various automata?**

**Answers for Exercise 2.**

**(1) Words that end with  $ab$ .** I use three states that remember the last one or two symbols. The DFA is:

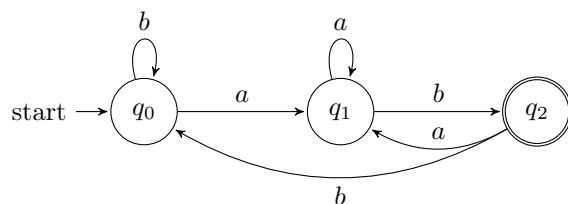


Figure 1: DFA for all words that end with  $ab$ .

**(2) Words that contain  $aba$ .** Here I track how much of the pattern  $aba$  I have just seen as a suffix.

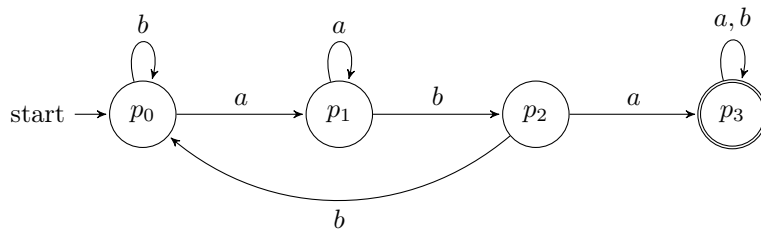


Figure 2: DFA for all words that contain  $aba$  as a substring.

**(3) Odd number of  $a$ 's and odd number of  $b$ 's.** For this and the next language I use the same 4-state parity automaton, but choose different accepting states. The four states remember the parity (even/odd) of the number of  $a$ 's and  $b$ 's seen so far.

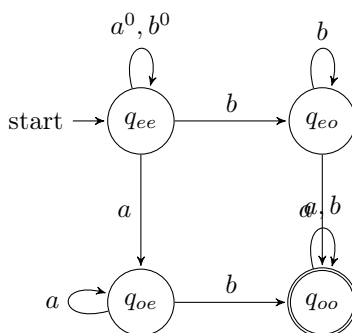


Figure 3: Parity DFA: state  $q_{xy}$  means  $x$ -parity for  $a$ 's and  $y$ -parity for  $b$ 's. For language (3) we accept  $q_{oo}$  (odd  $a$ 's and odd  $b$ 's).

**(4) Even number of  $a$ 's and odd number of  $b$ 's.** Using the same parity DFA in Figure 3, this time the accepting state is  $q_{eo}$  (even  $a$ 's, odd  $b$ 's) instead of  $q_{oo}$ .

**(5) Any three consecutive characters contain at least one  $a$ .** This is the same as saying that we never see  $bbb$  as a substring. I track runs of consecutive  $b$ 's up to length 3; hitting three  $b$ 's in a row moves to a dead (rejecting) state.

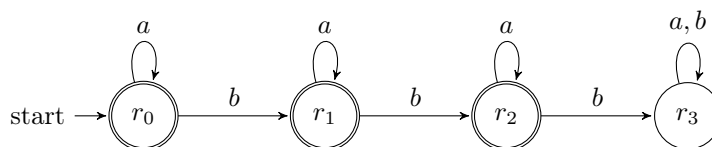


Figure 4: DFA that forbids  $bbb$ ; states  $r_0, r_1, r_2$  are accepting,  $r_3$  is a dead state reached after three  $b$ 's in a row.

**(6) Words that contain  $bbb$ .** I reuse the same transition structure as in Figure 4, but now  $r_3$  is the *only* accepting state (once we have seen three consecutive  $b$ 's we stay in an accepting sink).

**What I notice.** Several of these automata reuse the same underlying structure but just change which states are accepting (for example (3) vs. (4), and (5) vs. (6)). In other words, a lot of the “different” languages here really come from the same core DFA with a different choice of final states.

### Question

How might you combine the languages  $L(\mathcal{A}_1)$  and  $L(\mathcal{A}_2)$  from Exercise 1 using set operations (union, intersection, complement)? Could you represent the logic with boolean logic exclusively?

## 2.2 Week 2: Operations on Automata

### Product automata

#### Exercise 1 (Product automata)

Consider the following automata  $\mathcal{A}^{(1)}$  and  $\mathcal{A}^{(2)}$ .

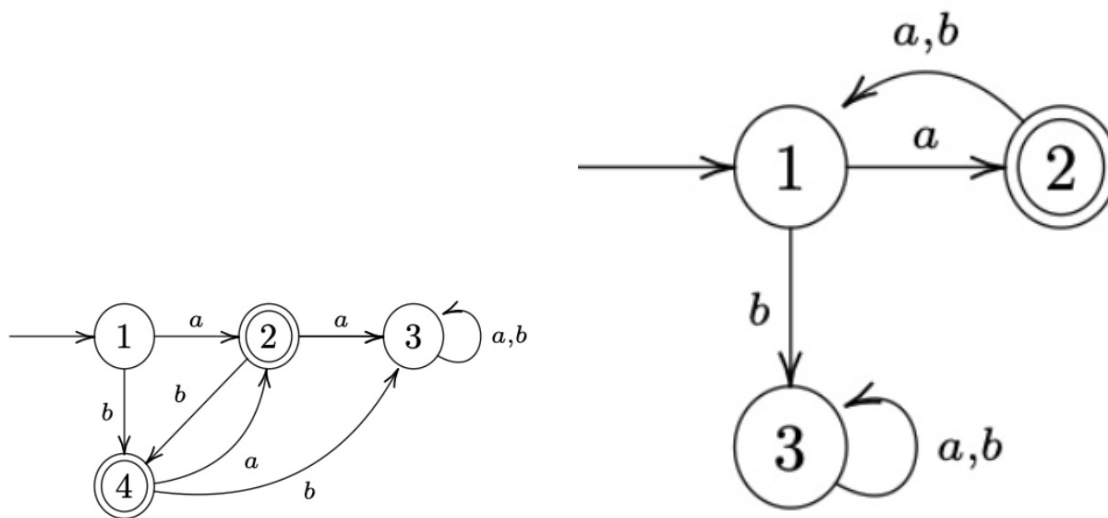


Figure 5:  $\mathcal{A}^{(1)}$  (left) and  $\mathcal{A}^{(2)}$  (right).

1. Describe  $L(\mathcal{A}^{(1)})$  and  $L(\mathcal{A}^{(2)})$ .

**Answer:**

- $L(\mathcal{A}^{(1)})$ : Words include  $a, b, ba, bab, baba, ab, aba, abab$ , etc. I can see that each character must alternate based on its successive character.
- $L(\mathcal{A}^{(2)})$ : Words include  $a, axa$  (where  $x$  doesn't matter). I see the word must start with  $a$ , and must add to its word length in increments of 2, every other character being  $a$ .

2. Construct the intersection automaton  $\mathcal{A}$  for  $\mathcal{A}^{(1)}$  and  $\mathcal{A}^{(2)}$  by drawing its transition diagram (omit unreachable states).

**Answer:** Below is the intersection automaton diagram.

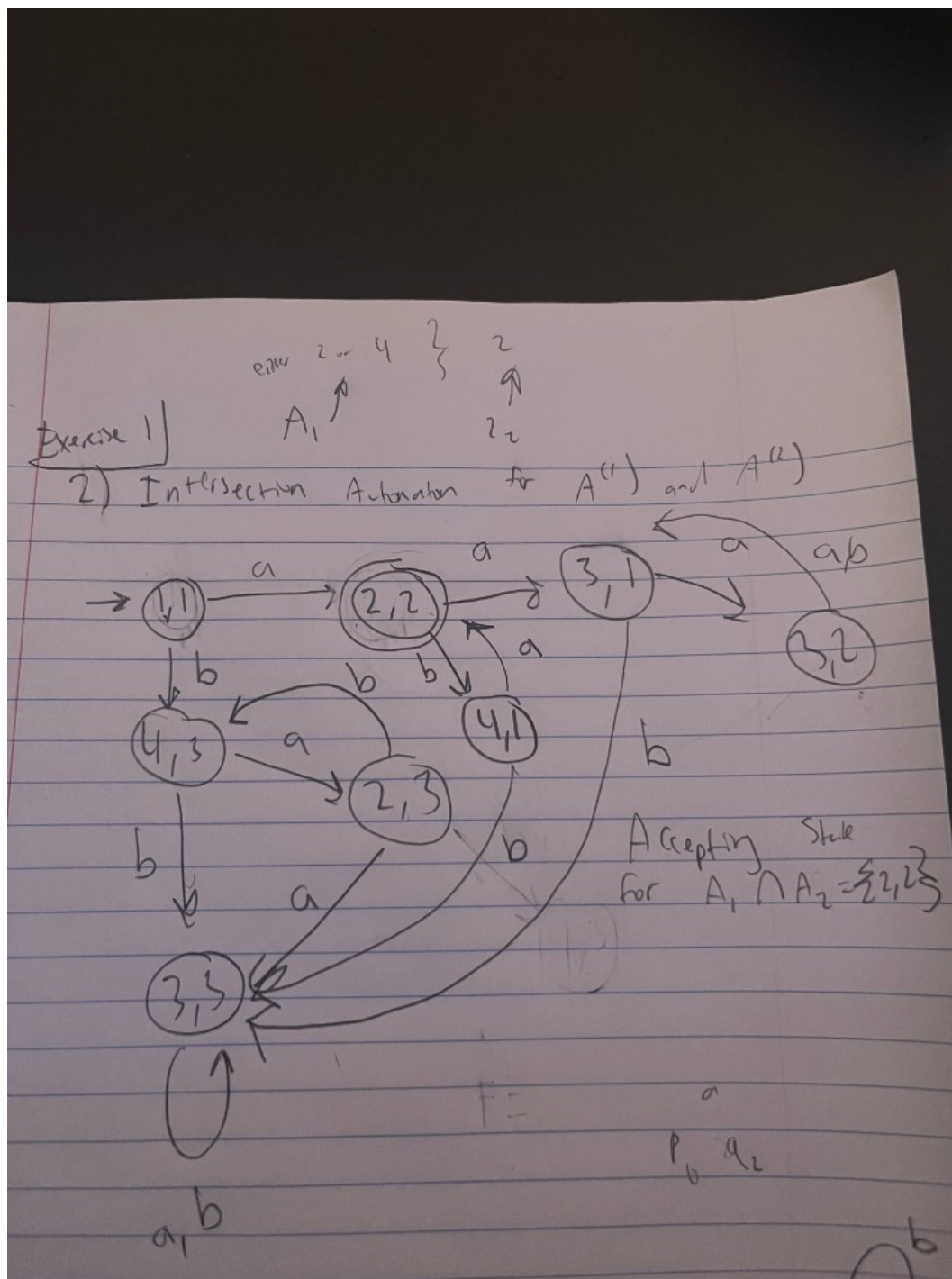


Figure 6: Intersection automaton  $\mathcal{A}$  for  $\mathcal{A}^{(1)}$  and  $\mathcal{A}^{(2)}$  (omit unreachable states).

Note:  $\{2, 2\}$  as accepting state for intersection product automaton.

3. Argue that  $L(\mathcal{A}) = L(\mathcal{A}^{(1)}) \cap L(\mathcal{A}^{(2)})$ .

**Answer:** Since  $L(\mathcal{A}^{(1)})$  includes words such as  $a$ ,  $b$ ,  $ba$ ,  $bab$ ,  $baba$  and  $ab$ ,  $aba$ ,  $abab$ , and  $L(\mathcal{A}^{(2)})$  includes words that (1) must start with  $a$ —so we can disregard the words that start with  $b$  in  $L(\mathcal{A}^{(1)})$ , leaving us with  $a$ ,  $ab$ ,  $aba$ , etc.—and (2) the requirement for  $L(\mathcal{A}^{(2)})$  is to be in increments of two, we are left with patterns:  $a$ ,  $aba$ ,  $ababa$ , etc. Thus we obtain the automaton of both intersection.

4. **How can you change  $\mathcal{A}$  to get an automaton  $\mathcal{A}'$  with  $L(\mathcal{A}') = L(\mathcal{A}^{(1)}) \cup L(\mathcal{A}^{(2)})$ ?**

**Answer:** In the product automaton we would add the accepting states to not be exclusive to both, but inclusive of either one of the subset DFA's accepting state.

### Exercise 2 (More automata)

Consider the following automata  $\mathcal{B}^{(1)}$  and  $\mathcal{B}^{(2)}$ .

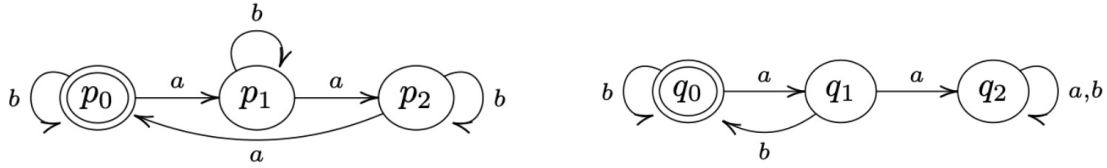


Figure 7:  $\mathcal{B}^{(1)}$  (left) and  $\mathcal{B}^{(2)}$  (right).

(Recall: the complement of a language  $L \subseteq \Sigma^*$  is  $\bar{L} := \Sigma^* \setminus L$ .)

1. **Describe  $L(\mathcal{B}^{(1)})$  and  $L(\mathcal{B}^{(2)})$ .**

**Answer:**

Figure 8:  $L(\mathcal{B}^{(1)})$  rules: include 3  $a$ 's at a time, any amount of  $b$ 's.  $L(\mathcal{B}^{(2)})$  rules:  $b$  must follow an  $a$ , any amount of  $b$ 's. DFA diagram for language 2.

2. **Construct the intersection automaton  $\mathcal{B}$  for  $\mathcal{B}^{(1)}$  and  $\mathcal{B}^{(2)}$  (omit unreachable states).**

**Answer:** Below is the intersection automaton diagram.

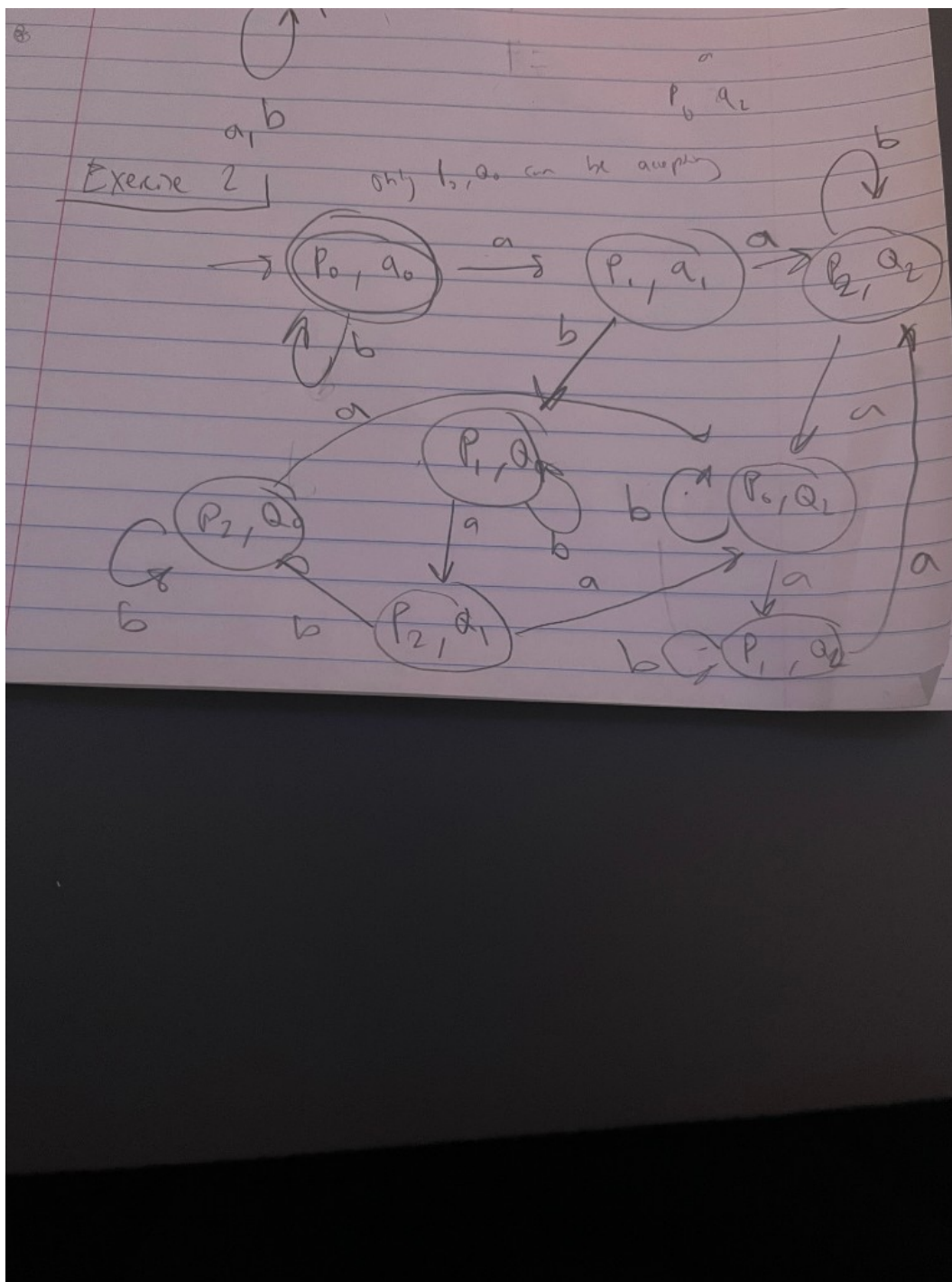


Figure 9: Intersection automaton  $\mathcal{B}$  for  $\mathcal{B}^{(1)}$  and  $\mathcal{B}^{(2)}$  (omit unreachable states).

Note:  $(P_0, Q_0)$  as accepting state.

3. Argue that  $L(\mathcal{B}) = L(\mathcal{B}^{(1)}) \cap L(\mathcal{B}^{(2)})$ .

**Answer:** Both automata have the rule that any amount of  $b$ 's is allowed. But we must combine the rules regarding  $a$ 's.  $L(\mathcal{B}^{(1)})$  requires 3  $a$ 's to be included at a time per word, while  $\mathcal{B}^{(2)}$  requires the rule that  $a$  must be followed by a  $b$ . This gives us the conclusion that words that do include an  $a$  must come in pairs of 3  $a$ 's with  $b$  following the last  $a$ .

4. **How can you change  $\mathcal{B}$  to get an automaton  $\mathcal{B}'$  with  $L(\mathcal{B}') = L(\mathcal{B}^{(1)}) \cup \overline{L(\mathcal{B}^{(2)})}$ ?**

**Answer:** The Union is equivalent to its intersection, no possible change.

### Exercise 3

Let  $\mathcal{A}$  be a DFA and  $q$  a state of  $\mathcal{A}$  such that from  $q$ , every input symbol  $a$  leaves us in  $q$  (i.e.  $\delta(q, a) = q$  for all  $a$ ). Show by induction on the length of the input that for any input string  $w$ , we still end up in  $q$  (i.e.  $\delta^*(q, w) = q$ ).

**Answer:** We show it by induction on the length of  $w$ . (Here  $\delta^*(q, w)$  is just the state we end up in after starting at  $q$  and reading the whole string  $w$ .)

When the input has length 0, the string is empty. So we don't read anything and we stay at  $q$ . So  $\delta^*(q, w) = q$  in that case.

Now suppose we've already shown that for every string of length  $n$ , we end up back at  $q$ . Take a string  $w$  of length  $n + 1$ . We can write it as  $w = va$  (some string  $v$  of length  $n$  plus one symbol  $a$ ). After reading  $v$  we're in  $\delta^*(q, v)$ , and by our assumption that's  $q$ . Then we read  $a$ , and we're in  $\delta(q, a)$ , which is  $q$  by the given rule. So after reading  $w$  we're still in  $q$ , i.e.  $\delta^*(q, w) = q$ .

So by induction, for all input strings  $w$  we have  $\delta^*(q, w) = q$ .

### Question

If you build the product automaton for  $\mathcal{B}^{(1)}$  and  $\mathcal{B}^{(2)}$  but change the accepting set to get the *union*  $L(\mathcal{B}^{(1)}) \cup L(\mathcal{B}^{(2)})$  instead of the intersection, how many states must be accepting?

## 2.3 Week 3

### Homework 1 (NFA)

1. Describe the language accepted by  $\mathcal{A}$  using a regular expression.
2. Specify the NFA  $\mathcal{A}$  in the form  $(Q, \Sigma, \delta, q_0, F)$ .
3. Find all paths in  $\mathcal{A}$  for the word  $w = abab$ ; represent in a diagram.

**Answer (NFA specification).** Here is how I write the NFA  $\mathcal{A}$  formally:

$$Q = \{0, 1, 2, 3\}, \quad \Sigma = \{a, b, c\}, \quad q_0 = 0, \quad F = \{2, 3\}.$$

For the transition function  $\delta : Q \times \Sigma \rightarrow \mathcal{P}(Q)$  I use

$$\begin{aligned} \delta(0, a) &= \{0, 1\}, & \delta(0, b) &= \{0\}, & \delta(0, c) &= \emptyset, \\ \delta(1, b) &= \{2\}, & \delta(1, c) &= \{2\}, \end{aligned}$$

and any transition not listed goes to the empty set (i.e. there is no move).



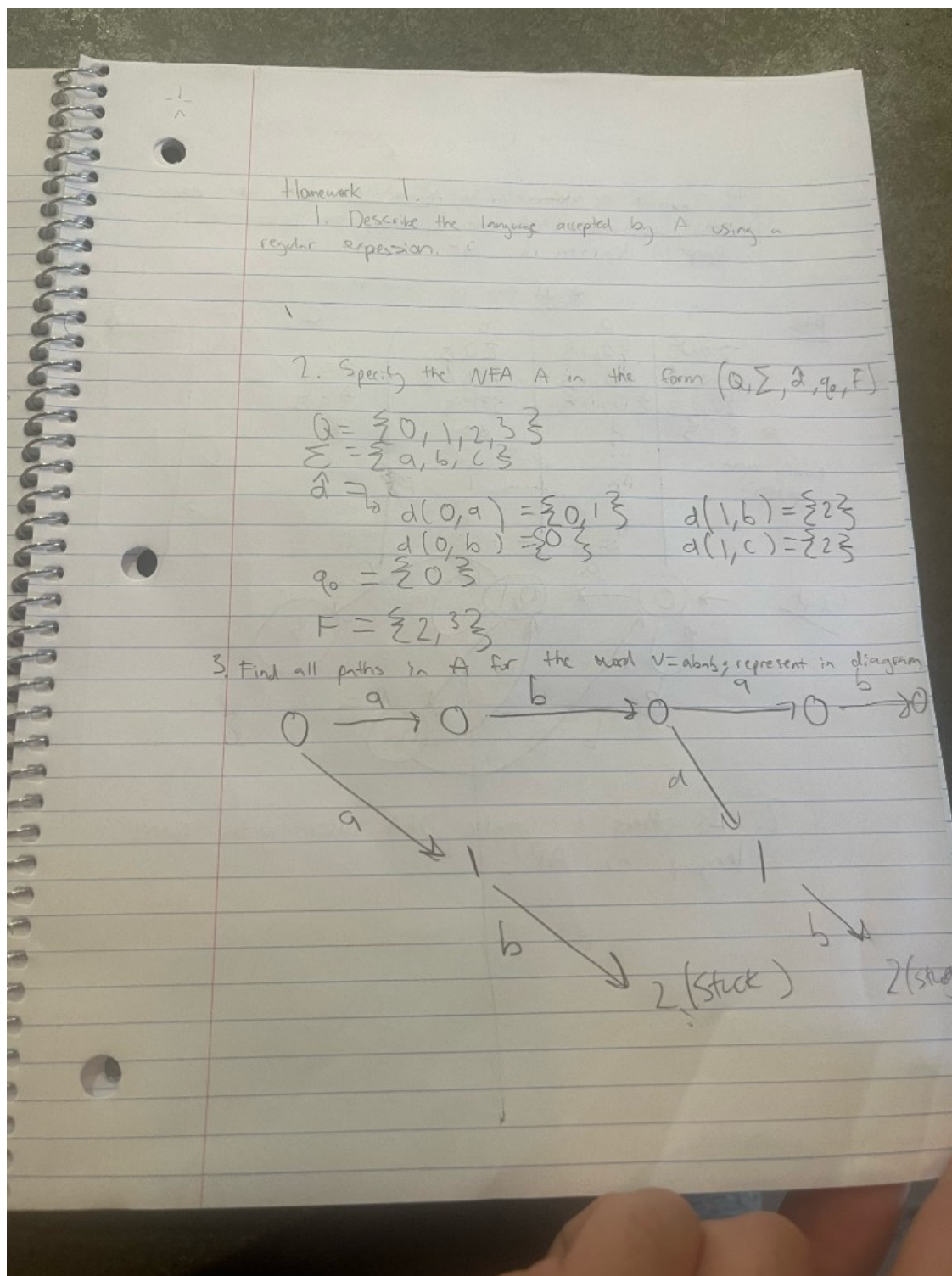


Figure 10: Homework 1: NFA specification  $(Q = \{0, 1, 2, 3\}, \Sigma = \{a, b, c\}, q_0 = 0, F = \{2, 3\})$  and all paths for  $w = abab$ .

## Homework 2 (Determinization)

Construct the determinization  $\mathcal{A}^D$  of  $\mathcal{A}$  using the power set construction. Create both a transition table and a graphical depiction of  $\mathcal{A}^D$ .

**Transition table for  $\mathcal{A}^D$  over  $\{a, b, c\}$ .** Starting from the NFA above, the reachable subset-states under the power set construction are

$$\{0\}, \{0, 1\}, \{0, 1, 2\}, \{0, 2, 3\}, \{2, 3\}, \emptyset.$$

I call a subset accepting when it contains either 2 or 3.

| State $S$     | on input $a$ | on input $b$ | on input $c$ |
|---------------|--------------|--------------|--------------|
| $\{0\}$       | $\{0, 1\}$   | $\{0\}$      | $\emptyset$  |
| $\{0, 1\}$    | $\{0, 1\}$   | $\{0, 2\}$   | $\{2\}$      |
| $\{0, 1, 2\}$ | $\{0, 1\}$   | $\{0, 2\}$   | $\{2\}$      |
| $\{0, 2\}$    | $\{0, 1\}$   | $\{0, 2\}$   | $\emptyset$  |
| $\{2\}$       | $\emptyset$  | $\{2\}$      | $\emptyset$  |
| $\emptyset$   | $\emptyset$  | $\emptyset$  | $\emptyset$  |

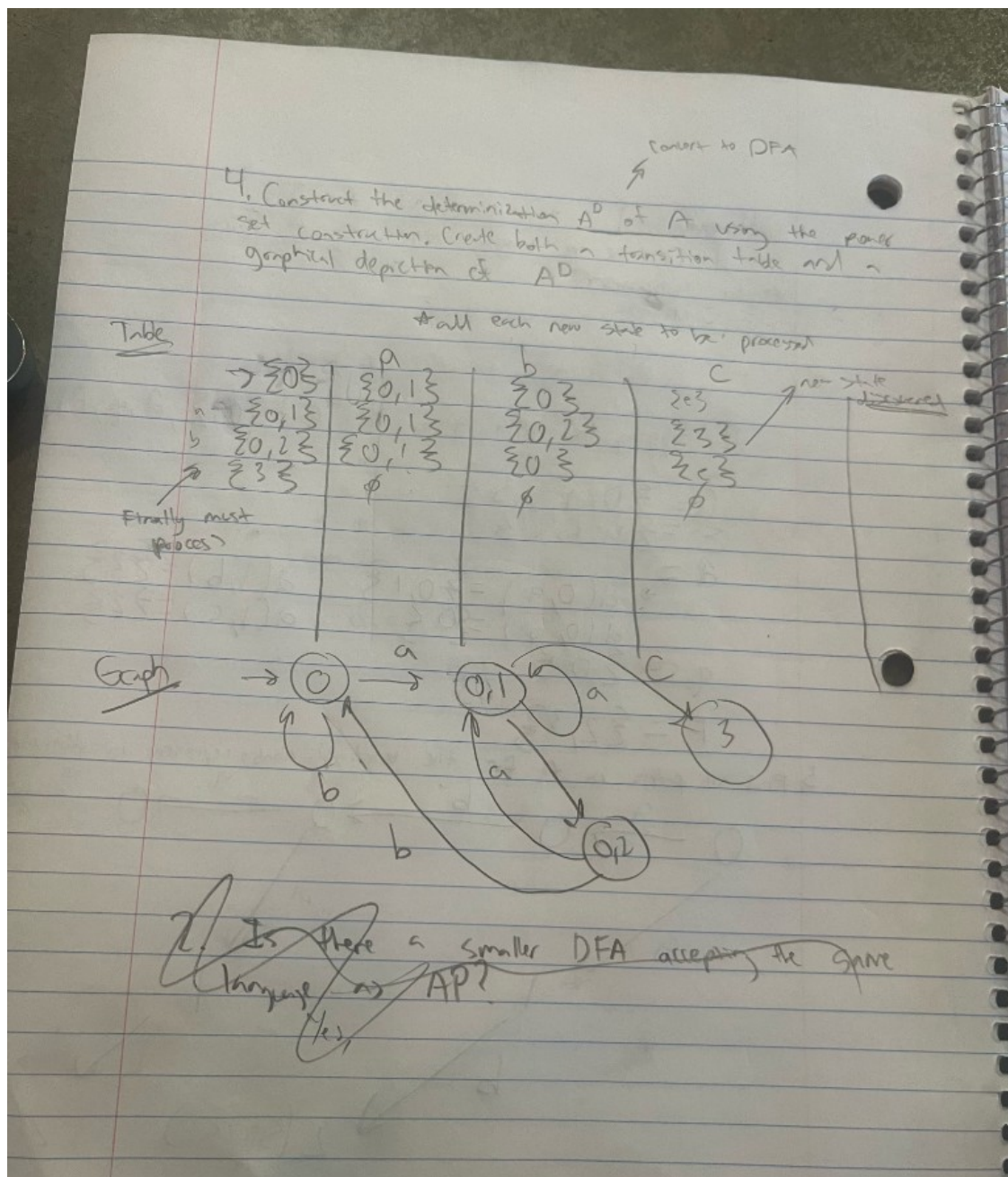


Figure 11: Graph of  $A^D$  (photo); the transition table is typeset separately in LaTeX.

### Homework 3 (DFA and minimization)

Determine  $A^D$  of  $A$  using the power set construction, now with alphabet  $\{0,1\}$ . Create both a transition table and a graphical depiction of  $A^D$ . Then decide whether there can be a smaller DFA accepting the same language.

**Transition table over  $\{0, 1\}$ .** From the NFA given in the homework, the reachable subset-states are

$$\{q_0\}, \{q_0, q_1\}, \{q_0, q_1, q_2\}, \{q_0, q_1, q_2, q_3\}, \{q_0, q_3\}.$$

The determinized DFA  $\mathcal{A}^D$  has the following transitions:

| State $S$                | on input 0               | on input 1     |
|--------------------------|--------------------------|----------------|
| $\{q_0\}$                | $\{q_0, q_1\}$           | $\{q_0\}$      |
| $\{q_0, q_1\}$           | $\{q_0, q_1, q_2\}$      | $\{q_0, q_1\}$ |
| $\{q_0, q_1, q_2\}$      | $\{q_0, q_1, q_2, q_3\}$ | $\{q_0, q_1\}$ |
| $\{q_0, q_1, q_2, q_3\}$ | $\{q_0, q_1, q_2, q_3\}$ | $\{q_0, q_1\}$ |
| $\{q_0, q_3\}$           | $\{q_0, q_1, q_2, q_3\}$ | $\{q_0, q_1\}$ |

Any subset that contains the original accepting state(s) is accepting in  $\mathcal{A}^D$ .

**Smaller DFA?** Yes. In this table the last two rows (for  $\{q_0, q_1, q_2, q_3\}$  and for  $\{q_0, q_3\}$ ) have exactly the same outgoing transitions and the same accepting/non-accepting status. That means these two states are equivalent and can be merged into a single state, giving a smaller DFA that still recognizes the same language.



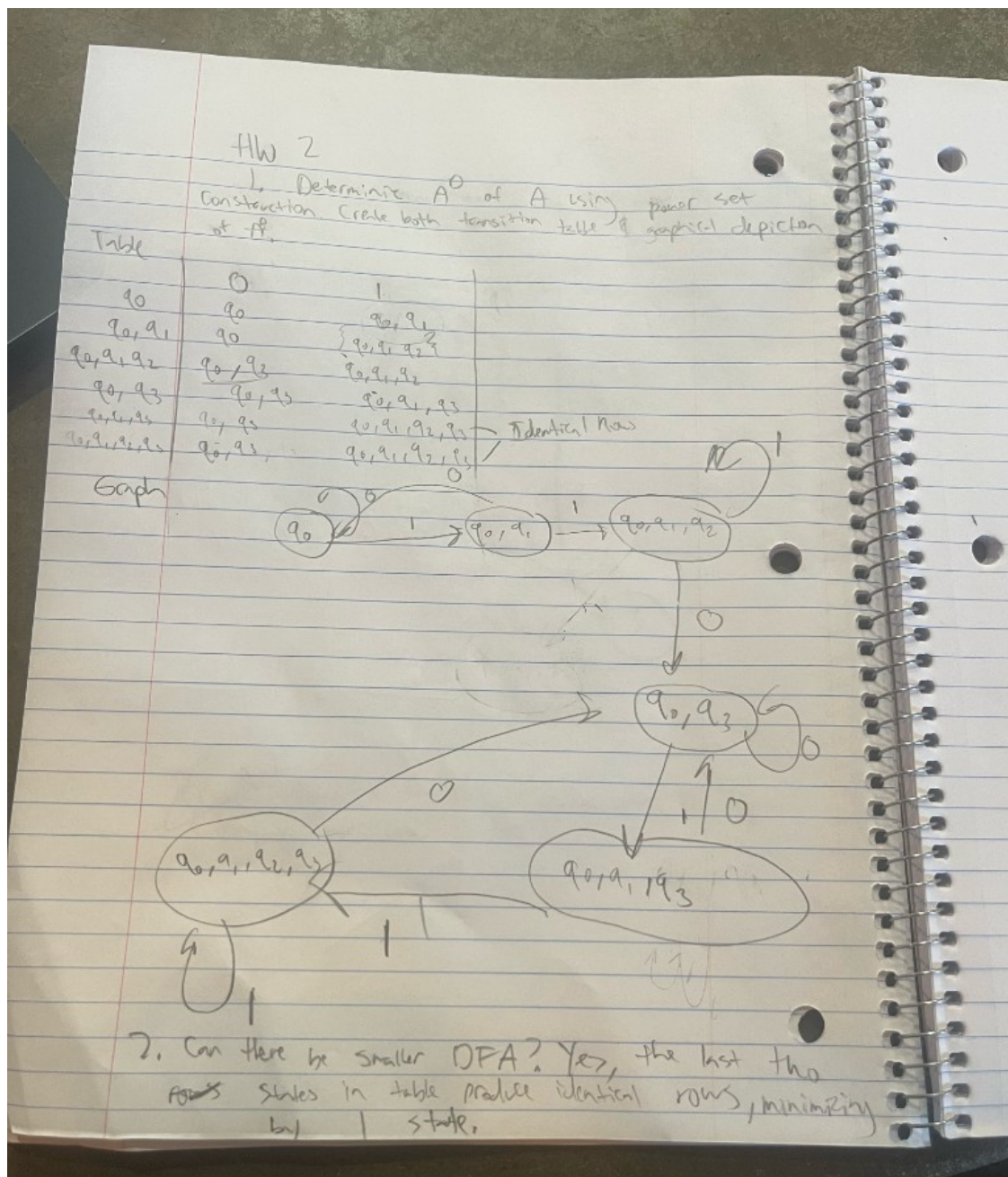


Figure 12: State diagram of  $A^D$  for Homework 3 (photo of the graph only).

### Question

After building  $A^D$  from an NFA  $A$  using the power set construction, when can we get a smaller DFA that still accepts the same language?

### **3 Synthesis**

### **4 Evidence of Participation**

### **5 Conclusion**

### **References**

[BLA] Author, [Title](#), Publisher, Year.